BCS714D - Big Data Analytics Assignment

Guidelines & Instructions

1. Track Overview & Evaluation Scheme

- Every group must choose **one track** (Track A or Track B) for their assignment submission.
- **All assignments require applying (the 3 core themes listed)** distributed computing, data management, and query benchmarking to your group's assigned dataset.

Track	Component	Weightage	Final Deliverable Requirements
Track A: Paper Presentation (Publication)	3 Themes + Benchmarking + Paper	**25 Marks**	1. Research Paper (IEEE Format) based **solely** on paper submission*. 2. Presentation & Demo.
Track B: Practical Engineering	**Part 1:** 3 Core Themes **Part 2:** 10-Mark Practical Extension	**15 Marks** **10 Marks**	1. Detailed Assignment Report covering **all** completed components (for both parts separately). 2. Presentation & Demo.

2. Core Themes (Mandatory for All Groups)

Every group must complete all 3 assignment themes on their assigned dataset.

For Sections 7A and 7B
Academic Year – 2026 – 2027
Faculty – Sandhiya B

Assignment Theme 1 — MongoDB Big Data Processing & Analytics

- **Document Data Ingestion:** Store, index, and organize large-scale structured/unstructured dataset into MongoDB collections.
- **Aggregation Pipelines:** Implement queries using $match, $group, $project, $unwind, and $facet to perform analytical processing.
- **Query Performance Profiling:** Evaluate index performance and execution speed using .explain("executionStats").

Assignment Theme 2 — Hadoop File Management & MapReduce Analytics

- **HDFS Operations:** Install Hadoop and perform distributed file management tasks:
  - Creating directories and adding files (hdfs dfs -mkdir, hdfs dfs -put).
  - Retrieving files (hdfs dfs -get, hdfs dfs -cat).
  - Deleting files and directories (hdfs dfs -rm -r).
- **MapReduce Execution:** Program, compile, and execute MapReduce jobs (Java or Python Streaming) on YARN to analyze the stored dataset.

Assignment Theme 3 — Pig Latin & Hive RCFile Performance Analytics

- **Data Transformation in Pig:** Write a Pig Latin script to load, clean, and filter the raw structured dataset.
- **Hive Export & RCFile Format:** Ingest the filtered data into Apache Hive and store it using the RCFile (Record Columnar File) format alongside a Standard Text format table.
- **Performance Analysis:** Execute identical analytical Hive queries (HQL) against both formats to analyze and report performance differences in query execution time and disk compression.

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Track A - 25 Marks

- Track A focuses on treating your implementation across all 3 themes as a research experiment. Instead of submitting a standard assignment report, your final grade is evaluated through a formal conference research paper and presentation*.

Track B - Part 1 – 15 Marks

- Same as Track A but you will be submitting a report instead of paper.

Track B – Part 2 – 10 Marks

- Groups taking Track B must complete one practical extension (or a custom extension approved for their dataset statement):

- **Sample Option 1** (Cross-System Hybrid Integration): Connect MongoDB directly to Hadoop/Hive using native connectors to perform cross-platform batch querying and real-time processing.

- **Sample Option 2** (Performance Profiling & System Stress Testing): Benchmark execution times and resource consumption (CPU, I/O, Memory) across MapReduce, Pig, and Hive under varying cluster parameters (e.g., HDFS split sizes, block sizes, mapper/reducer counts).

- **Sample Option 3 (Web Dashboard & Visualization (Flask / UI)** – Building a front end or a data visualisation dashboard or using python flask.